

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Perform gray level operations images	
Experiment No: 7	Date:	Enrollment No: 92510133028

```

import cv2

import numpy as np

import matplotlib.pyplot as plt


# Read image directly in grayscale

src_image = cv2.imread('/content/1785219786630.jpg', cv2.IMREAD_GRAYSCALE)


# Check if image is loaded

if src_image is None:

    print("Image not found!")

    exit()


# -----

# 1. Image Negation

# -----

negative_image = 255 - src_image


# -----

# 2. Thresholding

# -----

_, thresholded_image = cv2.threshold(src_image, 128, 255, cv2.THRESH_BINARY)


# -----

# 3. Gamma Correction

# -----

gamma = 2.0

```

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```
# Normalize image to range [0,1]
```

```
normalized_image = src_image / 255.0
```

```
# Apply gamma correction
```

```
gamma_corrected_image = np.power(normalized_image, 1/gamma)
```

```
# Convert back to range [0,255]
```

```
gamma_corrected_image = np.uint8(gamma_corrected_image * 255)
```

```
# -----
```

```
# Display Images
```

```
# -----
```

```
plt.figure(figsize=(15,5))
```

```
plt.subplot(1,4,1)
```

```
plt.imshow(src_image, cmap='gray')
```

```
plt.title("Original")
```

```
plt.axis('off')
```

```
plt.subplot(1,4,2)
```

```
plt.imshow(negative_image, cmap='gray')
```

```
plt.title("Negative")
```

```
plt.axis('off')
```

```
plt.subplot(1,4,3)
```

```
plt.imshow(thresholded_image, cmap='gray')
```

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```
plt.title("Threshold")
```

```
plt.axis('off')
```

```
plt.subplot(1,4,4)
```

```
plt.imshow(gamma_corrected_image, cmap='gray')
```

```
plt.title("Gamma")
```

```
plt.axis('off')
```

```
plt.show()
```

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

# Get uploaded filename automatically
filename = list(uploaded.keys())[0]

# Read image in grayscale
src_image = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

# Check if image loaded
if src_image is None:
    print("Image not found!")
else:
    print("Image loaded successfully!")

# Image Negation
negative_image = 255 - src_image

# Thresholding
_, thresholded_image = cv2.threshold(src_image, 128, 255, cv2.THRESH_BINARY)

# Gamma Correction
gamma = 2.0
```

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```

normalized_image = src_image / 255.0
gamma_corrected_image = np.power(normalized_image, 1/gamma)
gamma_corrected_image = np.uint8(gamma_corrected_image * 255)

# Display
plt.figure(figsize=(15,5))

plt.subplot(1,4,1)
plt.imshow(src_image, cmap='gray')
plt.title("Original")
plt.axis("off")

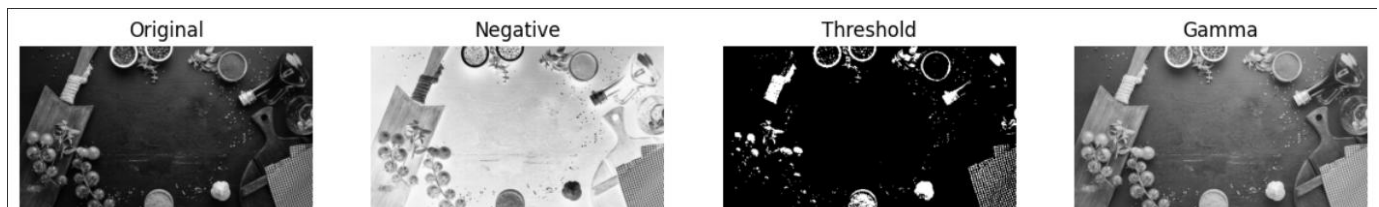
plt.subplot(1,4,2)
plt.imshow(negative_image, cmap='gray')
plt.title("Negative")
plt.axis("off")

plt.subplot(1,4,3)
plt.imshow(thresholded_image, cmap='gray')
plt.title("Threshold")
plt.axis("off")

plt.subplot(1,4,4)
plt.imshow(gamma_corrected_image, cmap='gray')
plt.title("Gamma")
plt.axis("off")

plt.show()

```



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All binary threshold img

```
# Different Thresholding Techniques

_, binary = cv2.threshold(src_image, 127, 255, cv2.THRESH_BINARY)

_, binary_inv = cv2.threshold(src_image, 127, 255,
cv2.THRESH_BINARY_INV)

_, trunc = cv2.threshold(src_image, 127, 255, cv2.THRESH_TRUNC)

_, tozero = cv2.threshold(src_image, 127, 255, cv2.THRESH_TOZERO)

_, tozero_inv = cv2.threshold(src_image, 127, 255,
cv2.THRESH_TOZERO_INV)

plt.figure(figsize=(18,10))

plt.subplot(2,3,1)
plt.imshow(src_image, cmap='gray')
plt.title("Original")
plt.axis("off")

plt.subplot(2,3,2)
plt.imshow(binary, cmap='gray')
plt.title("Binary")
plt.axis("off")

plt.subplot(2,3,3)
plt.imshow(binary_inv, cmap='gray')
plt.title("Binary Inverted")
plt.axis("off")

plt.subplot(2,3,4)
plt.imshow(trunc, cmap='gray')
plt.title("Truncate")
plt.axis("off")

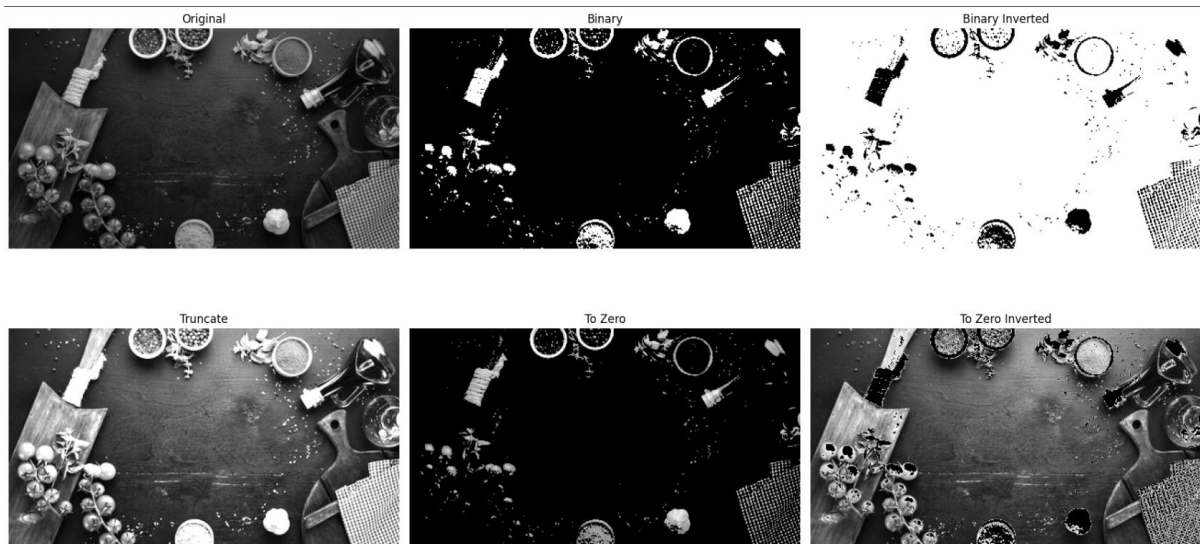
plt.subplot(2,3,5)
plt.imshow(tozero, cmap='gray')
plt.title("To Zero")
```

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```
plt.axis("off")

plt.subplot(2,3,6)
plt.imshow(tozero_inv, cmap='gray')
plt.title("To Zero Inverted")
plt.axis("off")

plt.tight_layout()
plt.show()
```



Different Gamma values

```
# Different Gamma Values
gamma_values = [0.5, 1.0, 2.0, 3.0]

plt.figure(figsize=(16,4))

# Original Image
plt.subplot(1,5,1)
plt.imshow(src_image, cmap='gray')
plt.title("Original")
plt.axis('off')

# Apply different gamma values
```

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```

for i, gamma in enumerate(gamma_values):
    normalized = src_image / 255.0
    gamma_corrected = np.power(normalized, 1/gamma)
    gamma_corrected = np.uint8(gamma_corrected * 255)

    plt.subplot(1,5,i+2)
    plt.imshow(gamma_corrected, cmap='gray')
    plt.title(f"Gamma={gamma}")
    plt.axis('off')

plt.tight_layout()
plt.show()

```



Conclusion

In this experiment, basic image processing operations were successfully performed using Python and OpenCV. Image negation produced the negative of the original image, thresholding converted the grayscale image into a binary image, and gamma correction adjusted the image brightness. The results demonstrate how these point processing techniques can enhance and transform digital images for different applications.

https://github.com/keshvi1234/DSIP_Exp